

Project RE-09-98 – Part I

Upper Yukon Radio Telemetry Tracking Station Installation
and
Spawning Ground Sampling / Tag Recovery

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Abstract

Remote satellite-linked tracking stations were established at the upstream and downstream limits of two areas surveyed annually by DFO to assess chum salmon abundance drainage-wide. These index areas are located on the mainstem Yukon River (between Tatchun Creek and Fort Selkirk) and on the upper Kluane River. The tracking stations operated effectively. Out of 530 chum salmon tags applied by U.S. NMFS at Rampart, Alaska, ten were tracked to and remained in each of the two index areas. The average rate of migration to the mainstem Yukon River index was 42 km/day and to the Kluane River index, 40 km/day. The radio-tagged fish which migrated to the mainstem Yukon River were captured at Rampart early and late in the four week long tagging operation. Those fish which migrated to the Kluane River were tagged late in the season. One mobile radio tag tracking survey was also conducted within each index area. No tags were identified as having passed the tracking stations undetected. No new spawning sites were identified within the index areas. Ground based surveys were conducted to document ratios of tagged fish to not-tagged fish (spaghetti tags), recover radio tags, and to collect age, size and sex information from the spawning escapement. Out of a sample of 600 fish, 0.5% had been tagged by DFO at the Canada/U.S. border (one of these fish exhibited a secondary mark applied by DFO but no tag) and 1.2% had been tagged by USFWS at Rampart. The sex ratio observed was 67% male; use of gillnets to capture fish for identification, age structures, and measurements, may have biased the sample towards males. Ageing structures were sent to Nanaimo for interpretation. No radio tags were recovered. Aerial surveys conducted by DFO identified approximately 7,300 chum salmon in each index area.

Key words: Yukon River, Canada, fall chum salmon, radio tags, satellite-linked tracking stations, spaghetti tags, mobile telemetry, spawning escapement.

Introduction

This report is intended to document the logistical and scientific aspects of the project for members of the Yukon Panel, various agencies, and the general public.

A need for better information on the distribution of chum salmon in the Yukon River drainage has been identified (see "Capabilities and Potential Applications of Adult Salmon Tagging Methods in the Yukon River Basin" by the Yukon River Joint Technical Committee, 1996). Radio telemetry has been established as an effective tool for providing distributional and behavioural information on salmon. Cursory radio telemetry studies on chum salmon were conducted in the Canadian section of the upper Yukon River in 1982 (Milligan et al, 1986).

Background on Rampart Tagging

Large scale spaghetti tagging studies were initiated at Rampart, Alaska in 1996. Feasibility studies and preparations for a large scale radio-telemetry study, to be conducted concurrently with the spaghetti tagging program, were also initiated. The objectives of the joint tagging studies include:

- estimation of the population of chum salmon migrating into the upper Yukon and Porcupine River drainages;
- determination of the relative distribution of these fish throughout these drainages, i.e. the stock composition of the return;
- apportionment of the run to country of origin; and
- identification of stock specific run timing and movement patterns.

Background and details of activities and results to date can be found in "Planning Team Proposal for Interagency Research in the Yukon River Drainage" by J.H. Eiler 1995, and in JTC (Canada/U.S. Yukon River Joint Technical Committee) 1996, 1997, and 1998.

A key component of the radio tagging study is a network of fixed, remote tracking stations which record the passage of radio-tagged salmon and transmit data to an office computer via satellite.

Fixed radio telemetry tracking stations were established at the following locations prior to 1998:

- eleven km upstream of the Rampart tagging site;
- close to the international border on both the mainstem Yukon¹ and Porcupine Rivers; and
- at the Fishing Branch River weir site.

Additional remote stations were established in 1998 prior to the chum salmon run. These stations include: the Sheenjek River, the Chandalar River, the Black River; the mainstem Porcupine River, (between the Sheenjek River and the Canada/U.S. border); and the mainstem Yukon River near Circle, Alaska (Figure 1).

¹ The specific location is the confluence of the Fortymile River.

In 1996, preparatory work for the studies included application of radio tags to fifty chum salmon at Rampart, Alaska in order to assess tagging response and to test equipment. Forty-eight (i.e. 98%) of the tagged fish were successfully tracked past stations located 11 km upstream of the tagging site.

In 1997, no radio tags were applied. Rather, effort was directed towards selecting and preparing potential tracking sites, as well as actual instalment of some stations.

In 1998, a total of 530 radio tags was applied to fall chum salmon at Rampart, Alaska. The tagging was conducted between August 24 and September 16. The same number of tags was applied during each of the four weeks of tagging. The radio tags used were manufactured by Advanced Telemetry Systems² and emitted signals in the 150 to 152 MHz range. Sixty (60) frequencies were used. The tags were pulse coded – each radio tag on a given frequency produced one of ten different signal codes. The tags were also equipped with activity monitors – if the fish was dead, or inactive for over eight hours, the signal pattern was altered.

Four hundred and seventy-one (471) of the radio-tagged fish moved upriver and were not caught in fisheries.

An automated database-GIS map, developed to assist in summarising telemetry data, was used effectively by U.S. NMFS during the 1998 season. Although a prototype, the system operated throughout the study and was essential in monitoring the remote tracking system, and the preparation of in-season summaries used to assess chum salmon returns and plan field operations. The GIS map includes both Canadian and U.S. sections of the Yukon River drainage. Additional work is being conducted to refine and enhance the database-GIS system capabilities. (JTC, 1998). It was not possible to create a web site for disseminating telemetry information during the 1998 season.

Study Area

The study area was as follows:

- the mainstem Yukon River between Tatchun Creek, and Fort Selkirk (which is just downstream of Pelly River confluence); and
- the Kluane River, between the Duke River and a short distance downstream of Kluane Wilderness Village.

These areas serve as abundance indices and are surveyed annually (using helicopters) by DFO.

² Mention of trade names does not constitute endorsement by DFO.

Objectives

This project had the following objectives for 1998, placed into two categories in accordance with methodology.

Category One Objectives:

- to record the number of chum salmon, radio-tagged at Rampart, Alaska, which migrate to two index spawning areas;
- to test the feasibility of using radio-telemetry for acquiring quantitative distributional information on chum salmon in the Yukon River;
- and to determine run timing and travel rates of radio-tagged fish.

Category Two Objectives:

- to locate spawning areas within indices which may be overlooked during visual inspection / enumeration
- identify tagged to not-tagged ratios (for spaghetti tags applied at the Canada/U.S. border, and at Rampart, Alaska);
- collect age, length and sex information from the spawning population;
- and to recover radio tags.

A major component of this project was the establishment of an infrastructure of fixed, remote radio tag tracking stations. This is intended to assist in the pursuance of longer term goals, which include the following:

- mapping of the current distribution of chum salmon spawning areas in the upper Yukon River drainage basin;
- identification of stock specific timing and migration patterns;
- evaluation of magnitude of the escapement to two index spawning areas relative to the upper Yukon River drainage wide escapement.

Note that achievement of the long term objectives is contingent upon application of sufficient radio tags in future years – the level of confidence placed in the results will be proportional to the number of radio tags applied.

Methods

The installation of four remote radio tag tracking stations at the upstream and downstream limits of two index areas in the upper Yukon River drainage were used in the pursuance of Category One

objectives. The methodology used in pursuance of the Category Two objectives included mobile tracking, visual tag searches, and sampling within the index areas.

Description of Remote Tracking Stations

The following was extracted from "A Remote Satellite-Linked Tracking Station System for Studying Pacific Salmon with Radio Telemetry" (Eiler, 1995), and is applicable to the tracking stations used in this project.

"Each station consisted of several integrated components, including a computer-controlled receiver developed by Advanced Telemetry Systems. The receiver and computer were used to detect, identify and record transmitter signals. The frequencies used for the radio-tagged fish were programmed into the computer. These frequencies were continually scanned by the receiver at 2-s intervals. When radio signals were detected, the frequency was scanned for 8-s, and the signals were analysed by the computer with a series of pattern-matching routines to determine whether a transmitter was present and to determine its code. Transmitter signals were recorded once per hour as long as they were in reception range.

Telemetry data – including frequency, code, data, and time that the transmitters were recorded, and activity (standard or inactive mode)-were stored by the computer....This information could be accessed and viewed directly from the computer's display screen or downloaded at the site to a portable computer for subsequent analysis.

Telemetry data were also transferred hourly to a second computer (data collection platform), manufactured by Handar Inc. (Sunnyvale, California)....The information collected was transmitted ... to a geostationary operational environmental satellite (GOES) and relayed to a receiving station in Camp Springs, Maryland.

Power for the receiver and data collection platform was supplied by sealed lead-acid batteries charged by solar panels. The voltage for each battery was recorded hourly and transmitted with other information.

Station components were attached to a 10-m portable tower. Two four-element Yagi receiving antennas at the top of the tower were positioned about 90° apart to provide uniform reception across and along the river...Cables connecting the station components were encased in metal tubing to prevent damage from rodents and bears."

The frequency of data transmission to the GOES satellite was every once every hour. A total of six batteries was used per station, in order to allow the equipment to continue operating throughout the winter (when solar panels are less effective).

Pictorial representation of the tracking system as well as the data transfer system is presented in Figures 2 and 3.

Installation of the mobile tracking stations involved a number of stages. These included the acquisition and upgrading of telemetry equipment, selection of sites for tracking stations, application for land use permits and radio licences, shipment of equipment to Whitehorse, site preparation, equipment placement and station assembly.

Acquisition and Refurbishment of Equipment for Tracking Stations.

This aspect of the project was undertaken by John Eiler of the U.S. NMFS. Details are presented on the attached. Items purchased included 24 six volt batteries, four sets of solar panels, antennae, coax cable and tower components. Also, for each of the four stations, a data collection platform (DCP), receiver, and remote data logger (RDL) was upgraded and refurbished. The amount spent on purchasing, upgrading or refurbishing equipment in preparation for station deployment was approximately \$20,000 U.S..

Selection of Tracking Station Sites

On Sunday May 24, sites suitable for tracking stations were selected close to the upstream and downstream limits of the mainstem Yukon River aerial survey index area. A helicopter was chartered from Carmacks³. Passengers were John Eiler, Brendan Kneller of the Selkirk First Nation, and Ian Boyce.

There were several criteria involved in the site selection process. One criterion was proximity to the index area boundaries. Another was maximisation of the "view" from potential radio-tagged fish to the tracking station receiving antennae. Optimally, a fish would be in the "view" of the station for as long as possible. This would decrease the chances of the signal from a given fish being overlooked while the receiver was scanning different frequencies. The radio tags in the study were of such a frequency that the signal emitted was very directional. Material blocking the line of sight, such as trees, gravel bars, bluffs would have a negative impact on signal reception. Placement of the station on a bluff or cliff at the edge of the river, and along a straight section of river, generally would have a positive effect on signal reception. Consideration was also given to accessibility of the site, with a view to security of the equipment. Placement of stations in areas of frequent traffic could mean an increase in the risk of tampering. In addition, a line of sight from the tracking station tower to approximately 10 degrees above the horizon, in a southerly direction, was also a search criterion. This line of sight was necessary for the transmission of data from the tracking station to the satellite (GOES), which is situated above the equator.

A location for the upstream station was chosen approximately one kilometre downstream from Tatchun Creek (see Figure 4). There has been no documentation of chum salmon spawning between the location selected and Tatchun Creek. The site is on a bluff with excellent, unobstructed views upstream and downstream on the Yukon River and to the satellite, providing

³ In conducting activities detailed in this report, helicopter access to the Kluane River index involved the Trans North Helicopters base in Silver City on Kluane Lake, or post-tourist-season, in Haines Jct. Helicopter access to the mainstem Yukon River index involved the base in Carmacks. The helicopter used was a Bell 206B Jet Ranger.

excellent potential for reception of radio signals and data transmission to the satellite. Accessibility of the site may be an issue as it is within sight, albeit some distance away, of the Tatchun creek area sport fishery. However, it is felt that with placement of educational signage at the fishery, tampering will be unlikely. This site is referred to as the Tatchun site for the remainder of this report.

No suitable site was found at the precise downstream limit of the index area (Fort Selkirk). However, one similar calibre (in terms of signal reception and transmission capabilities) to that of the Tatchun site was located approximately 20 river kilometres (rkm) downstream from Fort Selkirk (see Figure 4). It is located on a cliff and is remote and easily accessible only by helicopter, so security concerns are minimal. Chum salmon spawning has been documented between this location and the index area boundary. The location selected is referred to as the Selkirk site for the remainder of this report.

A similar process was undertaken on Tuesday, May 26 to select sites at the upstream and downstream limits of the Kluane River index. A helicopter was chartered from Silver City. Passengers were John Eiler, Gerald Dickson of the Kluane First Nation (KFN) and Ian Boyce.

Instead of using a fixed-wing aircraft as indicated in the project proposal, a helicopter was used. This decision was made based on discussions with John Eiler after the final proposal was submitted. It became apparent that the selection of a suitable site was a critical step in the project, and that closer inspection than that which could be afforded by a fixed-wing aircraft was required.

As with the mainstem Yukon River index, sites were found at the upstream boundary of the index and a short distance downstream from the lower boundary of the index. The upstream site is situated on a bluff approximately one kilometre from the confluence of the Duke River (Figure 5).

Considerable effort was expended trying to find a downstream site which satisfied the criteria listed above. Due to the relatively wide flood-plain and braided nature of sections of the Kluane River, this was achieved with some difficulty. An adequate site was eventually found approximately 3.5 river kilometres downstream from the index boundary (Figure 5). There is potential for chum spawning in between the site selected and the index boundary.

The upstream site on the Kluane River index is referred to as the Duke site in this report; likewise, the downstream site is referred to as the Toshingerman site since it is close to Toshingerman Lake. Both of the Kluane River sites are in areas of minimal boat traffic.

At all selected sites, flagging tape was used to delineate areas for clearing of trees and brush to allow placement of the stations and helicopter landing.

Land Use Permit and Radio License Applications

On June 2, application was made to the Land Resources Division of Indian and Northern Affairs Canada to clear helicopter pads and areas for tracking station assembly for the Selkirk site, the

Toshingerman site and the Duke site. Application for land use at the Tatchun site was made on June 7 to the Little Salmon Carmacks First Nation (LSCFN), since it is located on settlement land.

Approval for land use at the Selkirk, Duke, and Toshingerman sites was granted by Indian and Northern Affairs Canada on June 19. LSCFN granted written approval to clear a helicopter pad and an area for the tracking station at the Tatchun site on June 26.

On June 22, application was made to Industry Canada for licence to "install and operate (a) radio station(s) in Canada." Verbal authorisation was granted on July 6; two weeks later the licenses were received.

Shipment of Telemetry Gear

In anticipation of approval for land use at all four sites, most of the requisite gear (excluding batteries) was shipped via marine ferry from the U.S. NMFS Auke Bay Laboratory in Juneau to Skagway. On June 10 it was picked up by a DFO employee and driven to Whitehorse. The equipment was stored in the DFO Divisional office prior to mobilisation. Contact had been made with Customs and Excise, Revenue Canada well in advance of shipment of the goods. Bringing the goods into Canada was facilitated by a broker.

The batteries were shipped by land from Fairbanks to Whitehorse on June 29.

A list of the gear shipped into Canada is presented in Appendix 1. Please note that the list details only one tracking station, and therefore quantities (including value) should be multiplied by four.

Site Preparation

Clearing of the Selkirk site was arranged through the Selkirk First Nation (SFN). George McGinty and Alex Joe were contracted to do the clearing. These two individuals were ferried by helicopter to the site on July 7 and were picked up on July 8, which is the date on which the tracking stations were installed. This reduced the helicopter time required.

The Tatchun site was cleared on or about July 3 using one DFO employee (Marcel Barrault) and another member of the SFN, Brendan Kneller. Significant effort was expended in attempting to hire a member of the LSCFN; however, most candidates were either busy on other jobs or could not be reached.

The Toshingerman site was cleared concurrently with tracking station assembly on July 7, as there was sufficient area for the helicopter to land and sling in gear. However, the amount of space was marginal so an alternate helicopter landing zone was established for future use.

The Duke site was accessed by boat on July 1. Ian Boyce, accompanied by Billy Johnston and Derek Johnston of the Kluane First Nation (KFN), cached some of the tracking station gear, and conducted limited clearing for helicopter landing and station assembly.

Equipment Placement and Station Assembly

John Eiler and an employee of the U.S. Fish and Wildlife Service (USFWS), Randy Brown, provided the technical expertise required for station assembly. Both individuals arrived in Whitehorse on July 5; John Eiler from Juneau, Alaska and Randy Brown from Fairbanks, Alaska.

Station installation involved assembly of all the tower components (receiving and transmitting antennae, solar panels, data collection platform, RDL, receiver, etc.), and connection of power source. It included orientation of the two receiving antennae per tower so that maximum coverage of upstream and downstream stretches of river could be realised, as well as orientation of the transmitting antenna towards the GOES.

A reference tag was placed within 500 m to three km of each site in order to allow for a continuous check on station performance. After assembly was complete, an on-site check was conducted to ensure the station was functioning properly; specifically, that data reception and transmission were taking place.

The following table summarises some aspects of the equipment and station assemblies. Personnel listed are in addition to John Eiler, Randy Brown and Ian Boyce.

Station	Date	Transportation	Personnel
Duke	July 6	Boat	Gerald Dickson (KFN)
Toshingerman	July 7	Helicopter	Gerald Dickson (KFN)
Tatchun	July 8	Helicopter	
Selkirk	July 8	Helicopter	Bernice Johnnie (SFN)

The Duke site had to be accessed again on July 8 by helicopter (done while en route from the Toshingerman station to Burwash Landing) due to a malfunctioning data collection platform (DCP), which is responsible for transmission of data to the satellite. The unit was replaced at that time.

As indicated in the table, it was possible to install the mainstem Yukon River stations on the same day.

Aerial Surveys

The DFO aerial enumeration survey of the Kluane River index was conducted by helicopter on October 26. The surveyors were Pat Milligan and Bill Waugh (both employees of DFO). After the survey was completed, the index was flown again by the same individuals, using two receivers, in order to identify radio tags. Prior to the flight an "H" antenna was mounted on the front of the helicopter.

Due to the type of landing gear on the helicopter, it was not possible at the time to mount a larger Yagi antenna which would have increased reception range. However, this did not appear to be an issue during the survey, in part because the index constitutes a relatively small area. The sixty tag frequencies that had been recorded by the tracking station at the Canada/U.S. border were programmed into the receivers. Each receiver scanned one half of the frequencies in order to reduce the chance of a tag being overlooked. The area between the downstream limit of the tracking station was also checked for spawners and radio tags.

The surveys of the Kluane River index area was originally scheduled for October 16. However, inclement weather and reports of turbid water conditions forced a delay.

Aerial enumeration of the chum salmon in the mainstem Yukon River index was conducted on October 15 by Ian Boyce and Pat Milligan by helicopter.

The mainstem index was also flown, using a Maule M7 out of Whitehorse, on October 28 by Ian Boyce and Pam Vust (a DFO intern). The Maule M7 is a fixed-wing aircraft -- Yagi antennae were mounted under each wing. The proposal called for a ground based survey; however, due to the problems with equipment availability, resulting from the delay in the Kluane River index aerial survey, an aerial search for tags was undertaken. One receiver continually scanned for half of the tags which had been recorded by the tracking station at the Canada/U.S. border. Another receiver was used primarily to identify tags which had passed the Selkirk site, but also scanned for the other half of the Canada/U.S. border tag group for a portion of the survey. The area between the downstream limit of the index and the tracking station was also checked for radio tags.

Ground Based Surveys and Sampling

Survey timing was determined by reviewing historical run timing and by maintaining communication with the individuals conducting aerial surveys. Areas containing significant numbers of spawning chum salmon were located using map points and through GPS co-ordinates provided by the aerial survey crews.

Surveys were conducted in the Minto area from October 16-18 and from October 23-25. An 18' Lowe river boat with a 50hp jet outboard motor was used to access sites between Fort Selkirk and Minto via the launch site at Minto Landing. Yukon Crossing was accessed via the launch site at Tatchun Creek. In the Kluane River index area, foot surveys were conducted on October 13, and on October 26-27. The initial survey of each index area concentrated on obtaining live/dead counts and tagged to not-tagged ratios. Second visits were made to areas of higher spawning concentrations in order to collect biological samples and to recover tags from post-spawning fish.

Jake Duncan led the ground survey crew in the Kluane River area. Ritchie Johnson and Orion Johnson of the KFN assisted Jake Duncan and provided updates on survey conditions throughout the study period.

In the Yukon mainstem index area, the crew was led by Mary Ellen Jarvis and assisted by Jake Duncan as well as Derek Edwards and Lazuras Simon of the SFN.

Ground Based Enumeration

Two or three surveyors wearing Polaroid sunglasses walked the water's edge. Either river right or river left was chosen, depending on which provided best vantage point. Walking the bank, each surveyor tallied counts independently. The results from each survey reach were compared only upon completion of the entire survey area. Counts of live, dead, and number of tagged and not-tagged chum salmon were recorded.

Tag Recovery

Spaghetti tags from the DFO and USFWS tagging programs (at the Canada/U.S. border and Rampart respectively) were recovered from post-spawn carcasses and the date, location, and tag number were recorded. Radio telemetry equipment was not available because it was being used for aerial tracking. The ground crew searched for grey spaghetti tags that would indicate radio-tagged fish.

Biological sampling

Where practical, samples were obtained from post-spawning carcasses. However, in order to obtain the sample size (500 fish) required to effectively characterise the age and size structure of the spawning escapement, a 50' x 5" multi-filament gillnet was employed in the capture of live post-spawn chum salmon. Care was taken to set the net away from concentrations of redds. Salmon were chased into the net by slapping the surface of the water with branches or by arm-waving. The catch was immediately examined and females were released if external characteristics (e.g. loose eggs, distended body cavity) suggested they had not completed spawning. Males deemed to be in prime spawning condition were also released.

All chum salmon captured and killed were sampled for pectoral fins, size (post-orbital hypural and fork length), and sex using standard methods. Scales (ten per fish) were taken from a portion of the chum sampled. Spaghetti tags were recorded and removed, and the fish were examined for secondary marks, and then for tag scars if a secondary mark was found. Potential secondary marks were hole punches in the operculum (on fish tagged at the Canada/U.S. border) and left pelvic fin clips (on fish tagged at Rampart, Alaska).

Scales and pectoral fins were sent to the DFO Pacific Biological Station in Nanaimo, B.C. for age interpretation.

Results

Tracking Stations

Ninety three radio tags were recorded by the Canada/U.S. border tracking station (JTC, 1998). The first tagged fish passed the station on September 9. Four radio tags were reported as having been recovered in the aboriginal fishery. Up to 64 radio-tagged chum salmon migrated to areas other than the mainstem Yukon River index or the Kluane River index to spawn.

Mainstem Yukon River Index

Data on individual tags from the index tracking stations are presented in Appendix 2.

A total of 15 radio-tagged chum salmon moved upstream past the Selkirk station into the mainstem Yukon River index. Five of these tags continued past the Tatchun site. Of the radio-tagged fish which moved upstream passed the Selkirk station and remained within the index area, the first one was recorded on September 30, and the last one was recorded on October 24. On average, it took mainstem Yukon River index chum salmon 30 days to move the 1,216 km from the tagging site at Rampart. The average rate of travel was therefore 42 km per day (S.E. = 6 km per day). The rate of travel ranged from 31 to 48 km per day.

Chum salmon which terminated their migration between the Selkirk and Tatchun stations were tagged throughout the four week tagging project at Rampart. Approximately equal numbers of fish which were tracked into the Kluane River index were captured in each half of the tagging project. The total number of fish radio-tagged each week at Rampart is presented in Appendix 3.

Kluane River Index

As observed in the mainstem Yukon River index, ten radio-tagged chum salmon passed the downstream tracking station and remained within the index area. No tags were detected passing upstream of the Duke station.

On average it took radio-tagged fish 36 days to move the 1,402 km from the tagging site at Rampart. The rate of travel averaged 40 km per day (S.E. = 4 km per day) and ranged from 32 to 43 km per day.

The chum salmon which terminated their migration between the Tosingerman and Duke stations were tagged only in the first or second week of the four week tagging program at Rampart.

Aerial Mobile Surveys

Mainstem Yukon River Index

The individual counts of the two enumerators obtained during the mainstem Yukon River index helicopter survey were within 18% of each other and averaged 7,292 chum salmon.

The fixed-wing telemetry survey identified all of the ten chum salmon radio tags that the tracking station data indicated were within the index. No tags were located between the downstream boundary of the index (Fort Selkirk) and the Selkirk tracking station.

A signal was picked up on frequency 150.742 MHz in the Ingersoll Islands area but was not decoded. No tags emitting signals on this frequency had been documented by the Selkirk station.

No previously undocumented spawning sites within the mainstem Yukon River index were identified. However, the fixed-wing tracking lacked the precision that helicopter tracking with Yagi antennae would have afforded. Also, without more than one survey over the course of the spawning period, it would have been difficult to determine if the association of a radio tag with a specific site was a result of a fish holding, spawning or drifting downstream.

Kluane River Index

The individual counts of the two enumerators obtained during the Kluane River index helicopter survey differed by 40% and averaged 7,337 chum salmon. Only one fish was observed between the downstream boundary of the index and the Toshingerman tracking station. No radio tags were tracked to this area.

Nine of the ten frequencies recorded by the tracking stations were identified by the mobile receivers. Individual pulse codes were not decoded; consequently, the tenth tag may have been overlooked because it was on the same frequency as another tag.

A signal was picked up on frequency 151.763 MHz. No tags emitting signals on this frequency had been documented by the Toshingerman station.

The aerial telemetry did not identify any spawning sites within the index which have not been observed through visual surveys.

Ground Based Surveys and Sampling

All results referenced in this section are presented in Appendix 4. Also in this section, all tags referred to are spaghetti tags unless indicated otherwise.

Mainstem Yukon River Index

Eleven discrete spawning areas within this index were surveyed for numbers of live, dead, and tagged chum salmon. Detailed results from the surveys are presented in Table 1. The total number of live chum salmon observed in all areas combined was 968 fish, of which 15 were tagged. A total of 191 carcasses was observed. One of these carcasses was tagged. Among the live fish observed, no distinction was made between chum tagged by DFO and USFWS since the tags used by both agencies were orange in colour. Tagged to not tagged ratios are therefore best derived from the biological sample where identification of agency tags was possible. Of the 460 fish examined in the biological sample, three (0.7%) were tagged with DFO tags. This includes one fish found with a left-opercular punch but no tag. A further six (1.3%) were marked with USFWS tags. Tag ratio data from the biological samples are presented in Table 3.

A total of 460 chum salmon was captured and sampled for age, sex, size characteristics. Sex ratios were determined by examining the catch data as opposed to the actual sample data, because some fish were determined to be in spawning condition and released. Overall, males comprised 70% of the catch (Table 5). The positive bias towards male chum salmon is likely influenced by the selectivity of the gill net, used in capturing the majority of the sample, towards that sex. Age results were not available at the time of writing.

Kluane River Index

The initial survey (conducted on October 13) was hampered by turbid water conditions. Three known chum salmon spawning areas were visited at this time but only one area (near Glacier Creek) provided even marginal conditions for observing fish from ground level. The observed count, averaged from the independent counts of three surveyors, was 39 chum salmon, one of which was tagged (Table 2). In the two remaining accessible areas, visibility was limited to the top 25 cm of the water column. Fish were observed surfacing but accurate counts could not be achieved.

Turbid conditions persisted throughout the study period. KFN surveyors were re-visited the index areas on two different occasions prior to the second ground reconnaissance and reported that survey conditions were marginal.

Tagged to not tagged ratios are presented from the biological sample in Table 4, Appendix 4. Of the 105 fish examined in the Glacier Creek area, one was marked with a USFWS spaghetti tag. None of the 35 chum examined in the "below Glacier Creek" slough were marked.

A total of 140 chum salmon was sampled for age, sex and size data near Glacier Creek. Of these, 77 were male (55 %) and 63(45%) were female (Table 6). Age results were not available at the time of writing.

Discussion and Recommendations

The tracking stations were installed successfully and functioned effectively. The fact that signals were received during aerial surveys on two frequencies that had not been recorded by the stations means little, primarily because the tags were not decoded. Decoding involves error checking and helps to ensure that the signal received is actually from a radio tag used in the study. The signals received could have been from another wildlife study (especially in the Kluane area) or other extraneous radio traffic. The only tag on frequency 151.763 MHz which was detected by the Canada/U.S border station was recorded by the Selkirk station, (and therefore almost certainly did not go to the Kluane River). Only one tag on frequency 150.742 MHz was detected at the border station. However, it is unlikely that this tag would have passed the Selkirk station without being detected because of the high quality of the site; the signal received during the aerial survey was probably not from a radio-tagged chum salmon.

Statistical consultations have been made in order to determine if the number of both radio and spaghetti tags observed in each area falls within an expected range based on the parameters of the 1998 Rampart population estimate, and the visual enumeration data from each index area. A preliminary analysis suggests that the number of radio tags does; however, a conclusion awaits a closer examination of the Rampart tag application data. Calculation of the expected range will assist in assessing the migrational success of radio-tagged Kluane River index fish and mainstem Yukon River index fish.

The radio tags tracked to the Kluane River index were applied early on in the tagging process at Rampart. This suggests that the Kluane River index chum salmon stocks migrate past Rampart ahead of some other upper Yukon River or Porcupine River stocks. No such trend was observed in the unweighted data with mainstem Yukon River index stocks. Conclusions regarding stock timing should be made cautiously in light of the small sample size.

The results of this study can be compared to those of the 1982 Canadian upper Yukon River study. In that study, 114 radio tags were applied to chum salmon live captured at a number of sites downstream of Dawson City. The fish were tracked using a Cessna 185 equipped with a Yagi antenna. Of the 114 radio-tagged fish, 68 were tracked into various tributaries or mainstem spawning locations. Of these, 14 (21%) went to the Kluane River index area, and 15 (22%) went to the mainstem Yukon River index.

In 1998, of the 89 radio-tagged fish which were detected by the tracking station situated at the confluence of the Fortymile River but were not captured in fisheries, ten (11 %) terminated their migration in the mainstem Yukon River index and, likewise, ten (11%) terminated their migration in the Kluane River index. Based on the radio tag data (which, again, should be viewed cautiously in light of the low numbers involved), these two index areas appeared of lesser importance in 1998 than in 1982.

Of the drainage-wide upper Yukon River (Canada) spawning escapement estimate in 1982 (32,000 chum), the aerial count at the Kluane index comprised 17% and the aerial count at the mainstem index comprised 3%.

The preliminary 1998 upper Yukon River (Canada) spawning escapement estimate is a range, specifically, 46,305 to 65,567 chum salmon. The aerial counts of both the mainstem Yukon River index and the Kluane River index comprised 16% and 11% of these preceding values respectively. Thus, based on this data it appears that the Kluane River index was of similar importance (in terms of numbers of spawning salmon) in 1998 as it was in 1982. The mainstem Yukon River index appeared to be of greater importance. It is possible that in 1998 low water levels may have enhanced countability on that index.

It is recommended that exploratory mobile tracking be undertaken each year that a radio tagging project is conducted. This, in concert with data from the remote tracking stations, should provide valuable distributional information on upper Yukon River chum salmon.

Acknowledgements

Thanks to John Eiler of the U.S. NMFS for his enthusiastic involvement and technical advice throughout the project, and Randy Brown for assistance in tracking station assembly. Thanks to Mary Ellen Jarvis and Jake Duncan for organizing and implementing the ground (and water!) based spawning area studies. Thanks also to Mary Ellen Jarvis for assistance in summarising data and writing sections of this report. For their participation in the project thanks also to Gerald Dickson, Billy Johnson, Derek Johnson, Orion Johnson, and Ritchie Johnson of the Kluane First Nation; to Brendan Knellar, Alex Joe, George McGinty, Derek Edwards, and Lazuras Simon of the Selkirk First Nation; and to the DFO staff involved.

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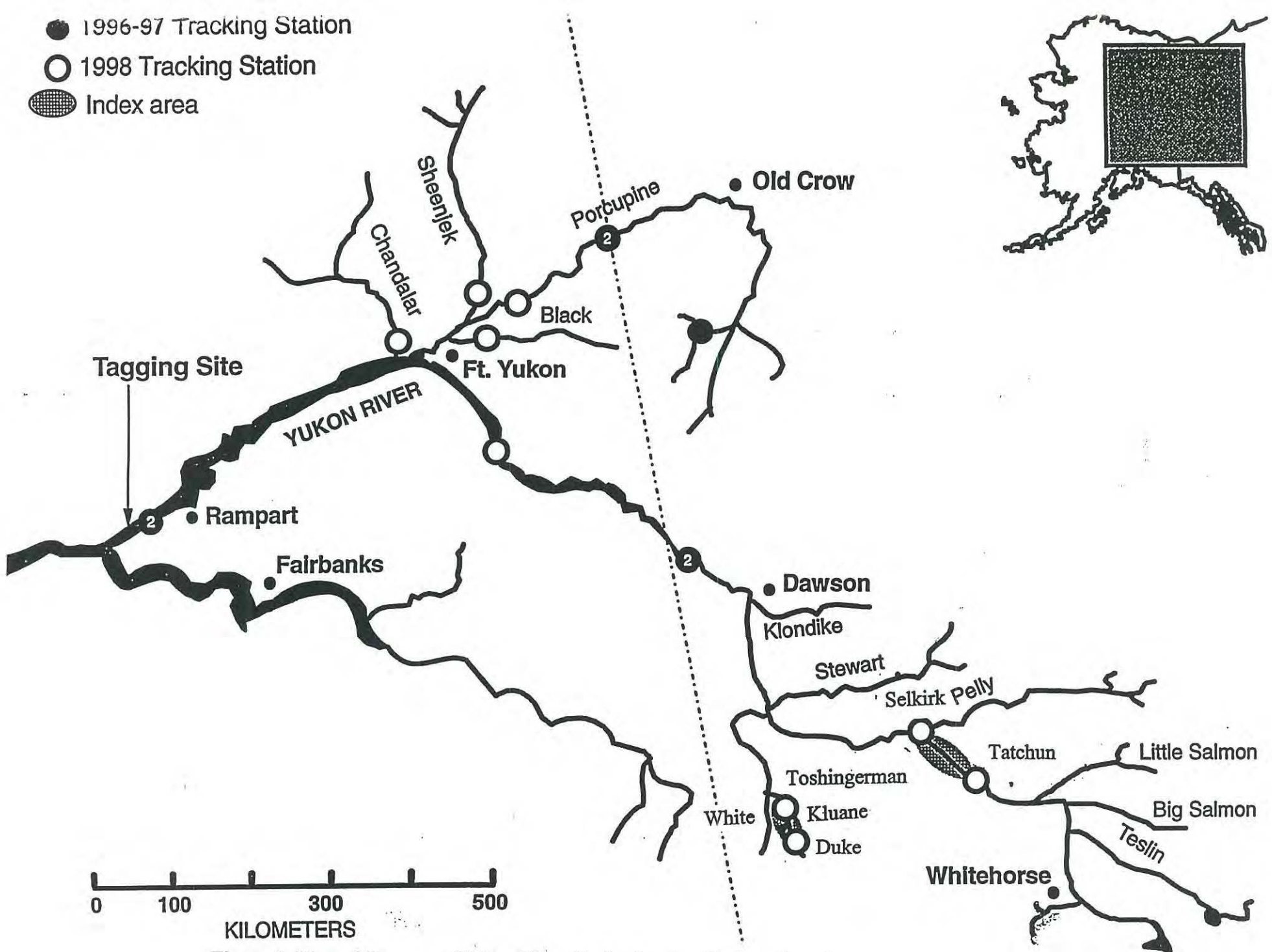


Figure 1. Map of the upper Yukon River Basin showing the location of remote tracking stations used to track radio-tagged salmon. Specific study areas are shaded.

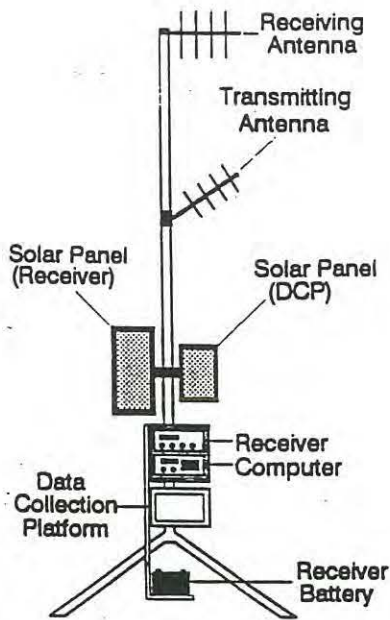


Figure 2. Remote tracking station used to detect and record salmon tagged with radio transmitters

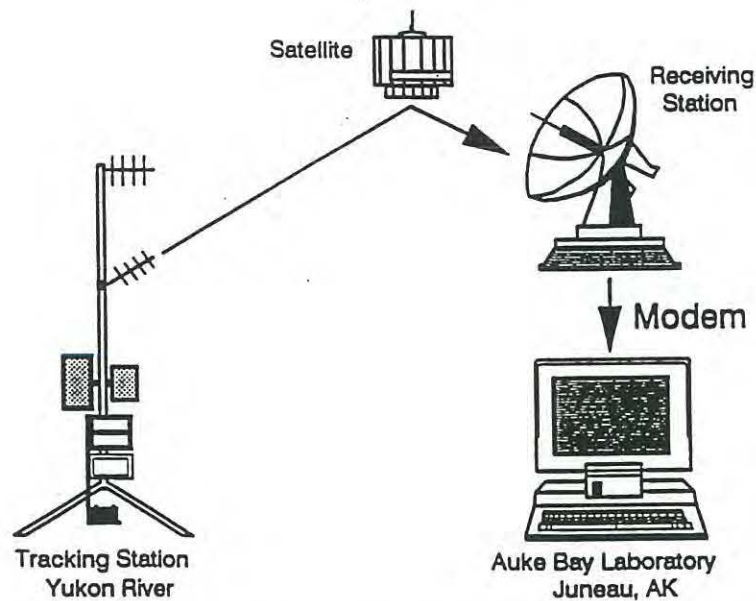


Figure 3. Data transfer system used to retrieve information collected by remote tracking stations



Figure 4. Mainstem Yukon River index and tracking stations.



Figure 5. Kluane River index and tracking stations.

Appendix 1. List of goods shipped to Canada for remote tracking stations.

INVENTORY LIST -- REMOTE TRACKING STATION COMPONENTS

<u>Item</u>	<u>Value (U.S.)</u>
Data collection platform (DCP)	5700.00
Radio receiver (150-154 MHz)	2700.00
Receiver data logger (RDL)	3100.00
Receiver/RDL Housing	700.00
Aluminum tower, portable	1520.00
GOES antenna	366.00
GOES antenna cable	110.00
Yagi antenna (3)	246.00
Coax cable, 20'	16.00
Coax cables, 3 (3)	27.00
Switch box	50.00
Battery box (fiberglass tote)	77.00
Battery cables	58.00
Battery regulator	110.00
Receiver batteries, 6 volt/210 amp hour (6)	1014.00
DCP battery	120.00
Solar panel, receiver (2)	904.00
Solar panel, DCP	330.00
SUBTOTAL (per station)	17,148

Appendix 2. Index area tracking station data.

Yukon River Fall Chum Program 1998 Telemetry Study

Chum Movement Rates

	Capture	Capture		Travel			Travel	Distance	Movement	
Fishno	Week ²	Date	Rapids RTS ³	Time	RTS Area	Date	Time (d)	(Km) ⁴	Rate ⁵	SE ⁶
106	1	8/24/98 20:00	8/25/98 12:00	0.67	Selkirk, Can MS	9/24/98 16:00	30.17	1216.4	40.32	
153	1	8/26/98 21:00	8/27/98 4:00	0.29	Selkirk, Can MS	9/21/98 13:00	25.38	1216.4	47.94	
179	1	8/27/98 15:00	8/27/98 23:00	0.33	Selkirk, Can MS	9/23/98 23:00	27.00	1216.4	45.05	
207	1	8/29/98 13:00	8/30/98 18:00	1.21	Selkirk, Can MS	9/26/98 11:00	26.71	1216.4	45.54	
221	2	8/31/98 11:00	8/31/98 23:00	0.50	Selkirk, Can MS	9/27/98 5:00	26.25	1216.4	46.34	
457	3	9/10/98 11:00	9/10/98 16:00	0.21	Selkirk, Can MS	10/6/98 11:00	25.79	1216.4	47.16	
561	4	9/14/98 11:00	9/14/98 20:00	0.38	Selkirk, Can MS	10/21/98 16:00	36.83	1216.4	33.02	
569	4	9/14/98 12:00	9/14/98 19:00	0.29	Selkirk, Can MS	10/15/98 1:00	30.25	1216.4	40.21	
595	4	9/15/98 11:00	9/15/98 18:00	0.29	Selkirk, Can MS	10/24/98 19:00	39.04	1216.4	31.16	
							29.71		41.86	6.19
154	1	8/26/98 21:00	8/27/98 6:00	0.38	Tatchun, Can MS	9/28/98 3:00	31.88	1324.4	41.55	
382	3	9/7/98 12:00	9/7/98 22:00	0.42	Tatchun, Can MS	10/8/98 7:00	30.38	1324.4	43.60	
479	3	9/10/98 12:00	9/10/98 18:00	0.25	Tatchun, Can MS	10/8/98 8:00	27.58	1324.4	48.01	
609	4	9/15/98 12:00	9/16/98 8:00	0.83	Tatchun, Can MS	10/25/98 3:00	38.79	1324.4	34.14	
622	4	9/16/98 11:00	9/16/98 18:00	0.29	Tatchun, Can MS	10/25/98 2:00	38.33	1324.4	34.55	
							33.39		40.37	5.98
101	1	8/24/98 19:00	8/25/98 1:00	0.25	Toshingerman (Kluane)	9/30/98 7:00	36.25	1402.7	38.70	
114	1	8/25/98 12:00	8/26/98 2:00	0.58	Toshingerman (Kluane)	10/4/98 13:00	39.46	1402.7	35.55	
176	1	8/27/98 15:00	8/27/98 22:00	0.29	Toshingerman (Kluane)	9/30/98 6:00	33.33	1402.7	42.08	
182	1	8/27/98 16:00	8/27/98 22:00	0.25	Toshingerman (Kluane)	10/1/98 8:00	34.42	1402.7	40.76	
183	1	8/27/98 16:00	8/28/98 1:00	0.38	Toshingerman (Kluane)	10/11/98 13:00	44.50	1402.7	31.52	
198	1	8/29/98 12:00	8/29/98 20:00	0.33	Toshingerman (Kluane)	10/3/98 11:00	34.63	1402.7	40.51	
235	2	8/31/98 14:00	9/2/98 5:00	1.63	Toshingerman (Kluane)	10/5/98 2:00	32.88	1402.7	42.67	
248	2	9/3/98 11:00	9/4/98 11:00	1.00	Toshingerman (Kluane)	10/7/98 23:00	33.50	1402.7	41.87	
261	2	9/3/98 12:00	9/3/98 21:00	0.38	Toshingerman (Kluane)	10/6/98 23:00	33.08	1402.7	42.40	
291	2	9/4/98 14:00	9/4/98 23:00	0.38	Toshingerman (Kluane)	10/8/98 22:00	33.96	1402.7	41.31	
							35.60		39.74	3.59
² Week1 from 8/23-29, Week2 from 8/30-9/5, Week3 from 9/6-12, Week4 from 9/13-19										
³ Remote tracking station (RTS) located 11.3 km upriver from Rapids tagging site										
⁴ Distance from Rapids RTS										
⁵ Kilometers per day										
⁶ Standard error for movement rate										

Appendix 3. Numbers of adult fall chum and chinook salmon tagged with radio transmitters at Rampart, Alaska by capture week during 1998.

Tagging Period (Capture week)	Fall Chum
Jul 19-25	
Jul 26-Aug 1	
Aug 2-8	
Aug 9-15	
Aug 16-22	
Aug 23-29	108
Aug 30-Sep 5	114
Sep 6-12	180
Sep 13-19	128
Sep 20-26	
Total	530

Appendix 4. Ground-based survey results.

Table 1. Survey results and tags observed: mainstem Yukon River index area

WP(i)	Site description	Duncan				Jarvis				Edwards				Average			
		live		dead		live		dead		live		dead		live		dead	
		count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged
106	Upper Yukon Crossing	244	2	6	0	244	2	5						244	2	6	0
107	Upper Yukon Crossing	229	2	0	0	202	2	0						216	2	0	0
109	Lower Yukon Crossing			1	0			1	0					poor survey conditions			
110	Yukon Crossing mainstem	76	0	1	1	81	0	1	1					79	0	1	1
111	LB above Ft Selkirk	17	1	0	0	19	1		0	17	0	0	0	18	1	0	0
111	LB above Ft Selkirk	95	3	0	0	85	3	0	0	77	3	0	0	86	3	0	0
111	LB above Ft Selkirk	62	1	0	0	54	0	1	0					58	1	1	0
111	LB above Ft Selkirk	294	4	4	0	286	7	4	0	314	5	6	0	298	5	5	0
112	RB above Pelly R.	23	1	0	0	21	2	1	0	18	2	1	0	21	2	1	0
112	RB above Pelly R.	23	0	0	0	26	0	0	0	24	0	0	0	24	0	0	0
112	RB above Pelly R.	10	0	0	0	12	0	0	0	12	0	0	0	11	0	0	0
113	RB above Pelly R.	18	0	0	0	20	0	0	0					19	0	0	0
114	Slough nr cabin, Rb	8	0	0	0	6	0	0	0					7	0	0	0
114	Slough nr cabin, Rb	18	0	0	0	14	0	0	0					16	0	0	0
115	Lower Ingersolls A	8	0	0	0	15	0	0	0	8	0	0	0	10	0	0	0
116	Lower Ingersolls B	103	0	4	0	95	0	6	0	100	0	2	0	99	0	4	0
117	RB above Ft.Selkirk.	211	4	175	0	234	4	175	0			175	0	223	4	175	0
	Totals	966	14	191	1	968	17	194	1	570	10	184	0	968	15	191	1

Note. All surveys were surveyed for tagged: untagged ratios during the first Minto survey with the exception of WP 117.

Way point 117 was surveyed on October 24.

Table 2. Survey results and tags observed: Kluane River index area

WP	Site description	Duncan				R. Johnson				O. Johnson				Average			
		live		dead		live		dead		live		dead		live		dead	
		count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged	count	tagged
na	Glacier Creek Slough	40	1	0	0	38	0	0	0	38	0	0	0	39	1	0	0
na	dns:poor survey conditions																
na	dns:poor survey conditions																
														39	1	0	0

Appendix 4. (Cont'd)

Table 3. Tag ratios observed in the biological sample, mainstem Yukon River index area.

WP	Site description	Examined	DFO Tags	USFWS Tags	% Tagged DFO	% Tagged USFWS
106	Upper Yukon Crossing	187	1	2	0.5	1.1
111	LB above Ft Selkirk	81	0	1	0.0	1.2
117	RB above Ft.Selkirk.	190	2	3	1.1	1.6
110	Yukon Crossing mainstem	1	0	0	0.0	0.0
109	Lower Yukon Crossing	1	0	0	0.0	0.0
	Totals	460	3	6	0.7	1.3

Table 4. Tag ratios observed in the biological sample, Kluane River index area.

WP	Site description	Examined	DFO Tags	USFWS Tags	% Tagged DFO	% Tagged USFWS
na	Upper Yukon Crossing	105	0	1	0.0	1.0
na	LB above Ft Selkirk	35	0	0	0.0	0.0
	Totals	140	0	1	0.0	0.7

Table 5. Sex ratio of fish caught by 5" gillnet in the mainstem Yukon River index area.

Wp	Location	sampled			released		sex ratio	
		#caught	males	female	males	female	%male	female
111	Lb above Ft Selkirk	85	57	24	0	4	67	33
117	Rb above Ft Selkirk	190	118	72	0	0	62	38
106	Upper Yukon Crossing	188	145	33	3	7	79	21
	Totals	463	320	129	3	11	70	30

Table 6. Sex ratio of fish caught by 5" gillnet in the Kluane Rive index area.

Wp	Location	sampled			released		sex ratio	
		#caught	males	female	males	female	%male	female
na	Glacier Creek	105	59	46	0	0	56	44
na	Below Glacier Creek	35	18	17	0	0	51	49
	Totals	140	77	63	0	0	55	45

(1) WP = way point ie geographical location



Photo 1. Selkirk tracking station on the mainstem Yukon River.



Photo 2. Tatchun tracking station on the mainstem Yukon River.



Photo 3. Toshingerman tracking station on the Kluane River.



Photo 4. Duke tracking station on the Kluane River.



Photo 5. Southern end of the mainstem Yukon River index. Tatchun Lake is visible in the background.



Photo 6. Spawning slough photographed during ground based surveys, mainstem Yukon River index.



Photos 7 and 8. Spawning ground sampling – taking structures for age determination.

Project RE-09-98 – Part II

PORCUPINE RIVER FALL CHUM TRACKING

Ian Boyce
Stock Assessment Biologist
Fisheries and Oceans Canada

FINAL REPORT
March 1999

Project RE-09-98 – Part II

Porcupine River Fall Chum Tracking

Ian Boyce, Stock Assessment Biologist, Fisheries and Oceans Canada

March 1999

Abstract

Aerial surveys were conducted on the Canadian section of the Porcupine River drainage using a Maule M7 aircraft fixed with two directional Yagi antennae. The purpose of the surveys, conducted in late October, 1998, was to gain distributional information regarding chum salmon which had been radio-tagged by the U.S. NMFS at Rampart, Alaska. Fixed remote tracking stations situated at the Canada/U.S. border on the mainstem Porcupine River and at the Fishing Branch River weir site were used to determine the number of tags in the study area. Of a total of 19 radio tags which moved into the Canadian section of the Porcupine River and were not harvested in the Old Crow area fishery or did not migrate to the Fishing Branch River, 11 were located. Nine of these tags were detected in the mainstem Porcupine River, and two were detected on the Miner River within 3 km of its confluence with the Fishing Branch River. At least five salmon, believed to be chum, were seen at one location on the mainstem Porcupine River in association with a radio tag signal. Apart from the Fishing Branch River, which was ice-free, the rivers surveyed were estimated to have at least 90% coverage by ice and snow. All radio signals detected were identified as being from tags in, or close to, open water areas.

Key words: Porcupine River, Canada, fall chum salmon, radio tags, aerial telemetry.

Introduction

This report is intended to document the logistical and scientific aspects of the project for members of the Yukon Panel, various agencies, and the general public.

A need for better information on the distribution of chum salmon in the Porcupine River drainage basin has been identified (see "Capabilities and Potential Applications of Adult Salmon Tagging Methods in the Yukon River Basin" by the Yukon River Joint Technical Committee, 1996). Radio telemetry has been established as an effective tool for providing distributional and behavioural information on salmon. Cursory radio telemetry studies on chum salmon were conducted in the Canadian section of the upper Yukon River in 1982 (Milligan et al, 1986). The areas of study did not include the Porcupine River drainage basin. The only area in the Porcupine River drainage basin which has been documented as supporting a significant amount of fall chum salmon spawning is on the Fishing Branch River.

The objective of this study was to gain information on fall chum salmon in the Porcupine River drainage basin in Canada with respect to:

- stock specific migration timing and movement patterns; and
- distribution of the spawning population.

This study involved tracking radio tags using a fixed-wing aircraft.

Background on Rampart Tagging.

Large scale spaghetti tagging studies were initiated at Rampart, Alaska in 1996. Feasibility studies and preparations for a large scale radio-telemetry study, to be conducted concurrently with the spaghetti tagging program, were also initiated. The objectives of the joint tagging studies include:

- estimation of the population of chum salmon migrating into the upper Yukon and Porcupine River drainages;
- determination of the relative distribution of these fish throughout these drainages, i.e. the stock composition of the return;
- apportionment of the run to country of origin; and
- identification of stock specific run timing and movement patterns.

Background and details of activities to date can be found in Eiler 1995, and in JTC (Canada/U.S. Yukon River Joint Technical Committee) 1996, 1997 and 1998.

allow full aerial coverage of the drainage basin, contributed to this decision. Documentation of movement patterns was placed at a lower priority than documentation of spawning distribution.

Consequently, an attempt was made to time mobilisation so as to be late enough that the bulk of the escapement to the Fishing Branch River had reached the weir site, but early enough that spawners would still be in the area within which they had spawned and had not drifted downstream a great distance.

Over the course of the summer, an investigation into aircraft availability and charter rates was conducted. A contract was negotiated with Alpine Aviation which involved a Maule M7 fixed-wing on wheels and skis. It had been anticipated at the time of proposal submission that a Cessna 185 would be used; however subsequent investigation revealed the merits of the Maule M7 in terms of charter rate, fuel economy and low stall speed (important for low level visual investigations of spawning areas).

A booking was made with Alpine Aviation of Whitehorse for October 19. On average, the run at the Fishing Branch River weir is complete by this time. Timing information provided by various stock assessment projects located throughout the Yukon River drainage suggested that the fall chum run to the upper part of the Yukon River drainage was one to two weeks late. Based on historical timing, if the run was one week late, the 1998 run would have been approximately 98% complete by October 19. Similarly, if it was two weeks late, the run would have been approximately 88% complete. By October 19, only about 22 spawned out chum salmon had drifted downstream onto the Fishing Branch River weir out of an escapement to date of approximately 13,000 fish. The weir is an estimated five river kilometres downstream of where the majority of Fishing Branch River chum salmon spawn. (In 1998, the Fishing Branch weir was pulled on October 22. The count for October 21 was 154 chum salmon – therefore, a few fish were still migrating at the time of demobilisation. By October 19, 96% of the fish that were counted had passed through the weir.)

The fixed tracking station at the Canada/U.S. border had recorded the passage of 36 radio-tagged chum salmon by October 19. The tracking station at the Fishing Branch River weir had picked up signals from 11 radio-tagged chum salmon. It was known that at least two radio tags had been captured in the aboriginal fishery in the vicinity of Old Crow. This meant that about 23 radio tags were unaccounted for in the Porcupine River drainage in Canada.

On the evening of Sunday, October 18, John Eiler arrived in Whitehorse from Juneau, Alaska. The following morning, a Yagi antenna was mounted under each wing of the Maule M7. Departure from Whitehorse was delayed until the next day due to extreme wind conditions which developed close to mid-day on Monday.

On Tuesday, October 20, the pilot (Gerd Mannsperger), John Eiler, and the proponent (Ian Boyce of DFO) left Whitehorse for Old Crow. A stop was made at the Dawson City airport

Porcupine River between Old Crow and the fixed tracking station, (a short distance downstream of the Canada/U.S. border), the Bluefish River, Lord Creek, the Old Crow River, and Thomas Creek. The following morning, the Driftwood River and the Bell River were surveyed. In the afternoon, Ian Boyce and Thomas Benjamin surveyed upper Thomas Creek, Timber Creek, Black Fox Creek and Johnson Creek.

On Sunday, October 25, John Eiler took a scheduled flight with Air North to Juneau via Fairbanks, Dawson City, and Whitehorse. Surveying was continued by Thomas Benjamin and Ian Boyce. Areas surveyed on the first excursion out of Old Crow included Lord Creek (which had been surveyed on October 23, but was en route to the area of interest), Cody Creek, the Miner River as far south as the point at which surveying ceased on October 20, the lower reaches of the Whitestone River, the Fishing Branch River (the south fork as far upstream as the weir site, and the entire north fork). After this circuit was completed, the aircraft returned to Old Crow where it was re-fuelled. Thomas Benjamin then remained in Old Crow while Ian Boyce flew in the direction of Dawson City. En route to Dawson City, the mainstem Porcupine River between Ellen Creek and the Miner River was surveyed. The lower five to ten kilometres of the Miner River was resurveyed; subsequently, the Whitestone River was flown to its headwaters.

Yukon Territory rivers not surveyed due to budgetary constraints include the Eagle River, Rock River and La Chute River. Also omitted for financial reasons was the area along the Canada/U.S. border south of the mainstem Porcupine River. This area includes the headwaters of the Salmon Trout River, Black River tributaries, and the headwaters of the Nation and the Kandik Rivers. Chum salmon entering these rivers would not have passed the tracking station situated at the border on the mainstem Porcupine River.

Results

Sixteen (16) of the 23 radio tags at large at the time of the surveys were identified. Three tags were heard in the town of Old Crow – these had obviously been recovered by fishers in the area. An additional two tags were observed in the vicinity of the Fishing Branch River weir; these tags were recorded by the tracking station on October 19. (One additional final tag was recorded by the tracking station at the Fishing Branch River weir on October 19, bringing the season total to 14. This tag was not detected during the aerial telemetry, perhaps because the fish may have moved to an area on upstream of the weir site which was not surveyed.) No radio tag signals were received on the Old Crow River or its tributaries, or the Bell River.

Table 1 presents radio tag information obtained, excluding tags detected in Old Crow or at the Fishing Branch weir site. Except for two signals identified on the Miner River², all signals received were on the mainstem Porcupine River (Figure 2). All of the tags were identified as being in, or immediately adjacent to, open water areas (except for those tags detected in the town of Old Crow). At the time of the survey, it is estimated that the

² The section of river between the Fishing Branch River outlet and the Whitestone River outlet is assumed to be the Miner River.

out of the water was much greater than that of submerged tags. The signal range was often over 10 kilometres for a tag out of the water. These tags could have been deposited by the current on a gravel bar or removed by predators such as bears or humans.

Visual verification of salmon presence was attempted at several locations at which tags were observed. Between Tizra Creek and Nukon Creek, the location at which tag 923-M12 was identified, at least five live salmon were observed. These fish appeared to be dark in colouration and were likely chum salmon. The lack of visual verification at the other locations does not mean that salmon were absent. Viewing was hampered by low light conditions, a seemingly dark river substrate, as well as low speed and low altitude limitations of the aircraft. Fish that were under snow and ice would not have been seen.

Some fishing occurs in the vicinity of the Bluefish/Porcupine River confluence. However, there were no nets in the water in that location at the time of the survey and the radio tag signal detected was relatively weak, indicating that the tag was submerged. An interview with the fisherman using the area indicated that he had not observed any grey spaghetti tags or radio tags in his catch. (Grey tags had been applied to the majority of fish which were radio-tagged at Rampart.) It is unlikely therefore that the radio tag detected close to the confluence of the Bluefish River and the Porcupine River was associated with a chum salmon which had been harvested.

Open water was observed in several locations in addition to those listed above. All of the mainstem Fishing Branch River that was seen was free of ice. (The reaches upstream of Bear Cave Mountain were not observed). The complete lack of ice on such a long stretch of river was in sharp contrast to the state of the other rivers surveyed. Open water was not observed on the north fork of the Fishing Branch River. Other open water areas documented include the lower few river kilometres of the Whitestone River and the Miner River, as well as isolated patches on these rivers, and the mainstem Porcupine River (including the base of Mason Hill). Ice-free areas were also documented on the Bell River, and Cody Creek at N66°33.34' / W138°46.11'. On tributaries to the Old Crow River, open water was observed at N68°33.71' / W140°01.72' and N68°27.96' / W129°49.20' (both of which are on the upper reaches of Timber Creek), as well as at N68°12.66' / W139°04.93 (which is on Black Fox Creek).

Discussion and Recommendations

The existence of chum salmon spawning at the locations in which radio tags were identified is not guaranteed. If a radio signal indicated that a given fish was alive and active, which was the case for four of the 11 radio signals identified in Table 1, it is possible that the fish was still migrating. Because of the timing of the surveys, however, the likelihood of this is considered low. A given inactivity radio signal received could have been associated with a fish that had drifted downstream from its spawning location. The distance that a fish would drift depends to a large extent upon the hydrology of the river in question. The radio tags listed in Table 1, however, had not passed the tracking

JTC (Canada/U.S. Yukon River Joint Technical Committee). 1997. Yukon River Salmon Season Review for 1997 and Technical Committee Report, 5-6 November, 1997. Whitehorse, Y.T.

JTC (Canada/U.S. Yukon River Joint Technical Committee). 1998. Yukon River Salmon Season Review for 1998 and Technical Committee Report, 18-19 November, 1998. Whitehorse, Y.T.

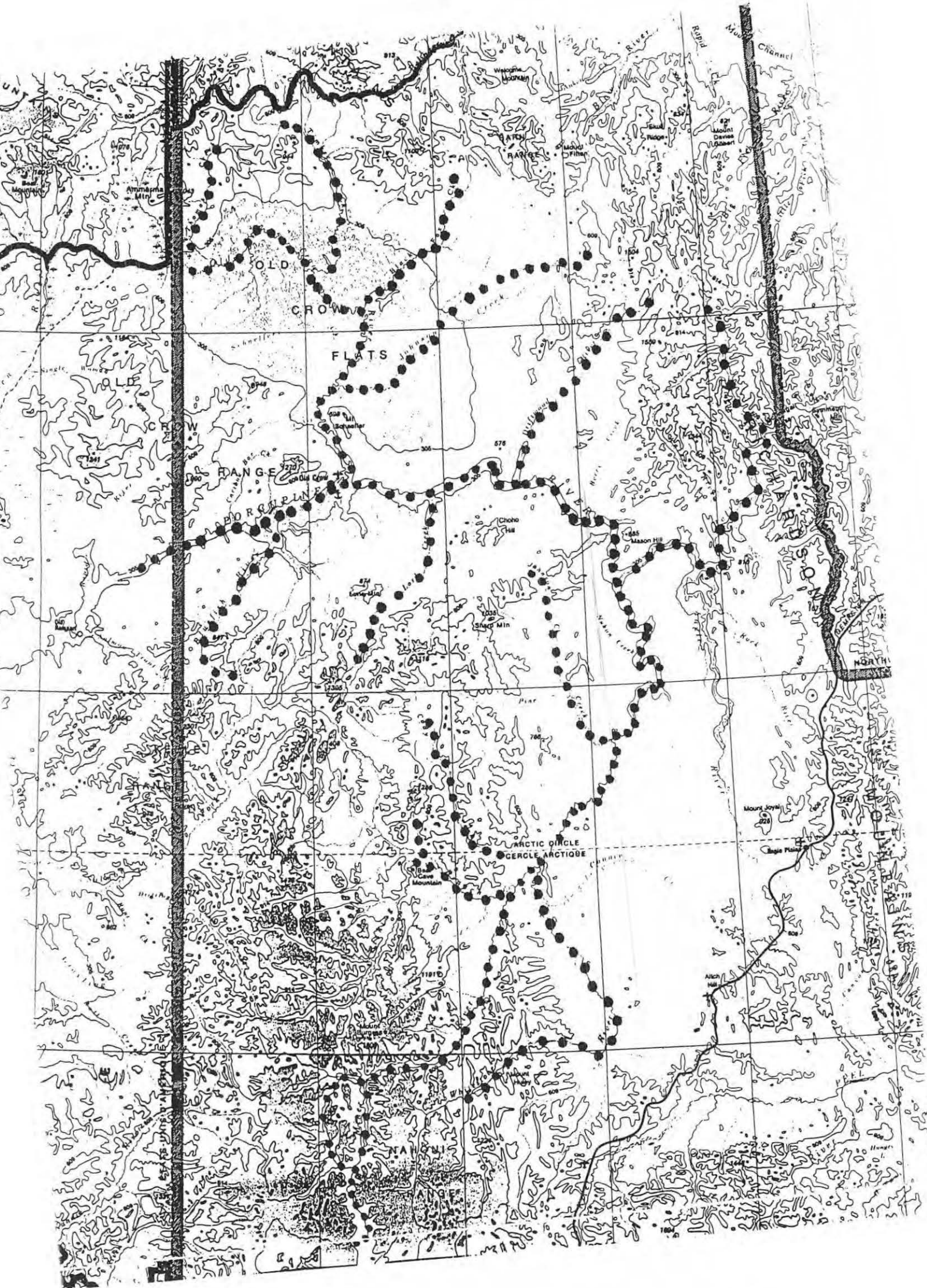


Figure 1. Porcupine River drainage in Canada. Areas surveyed are indicated with a dotted line. (Map - Yukon Territory 1:1,000,000.)

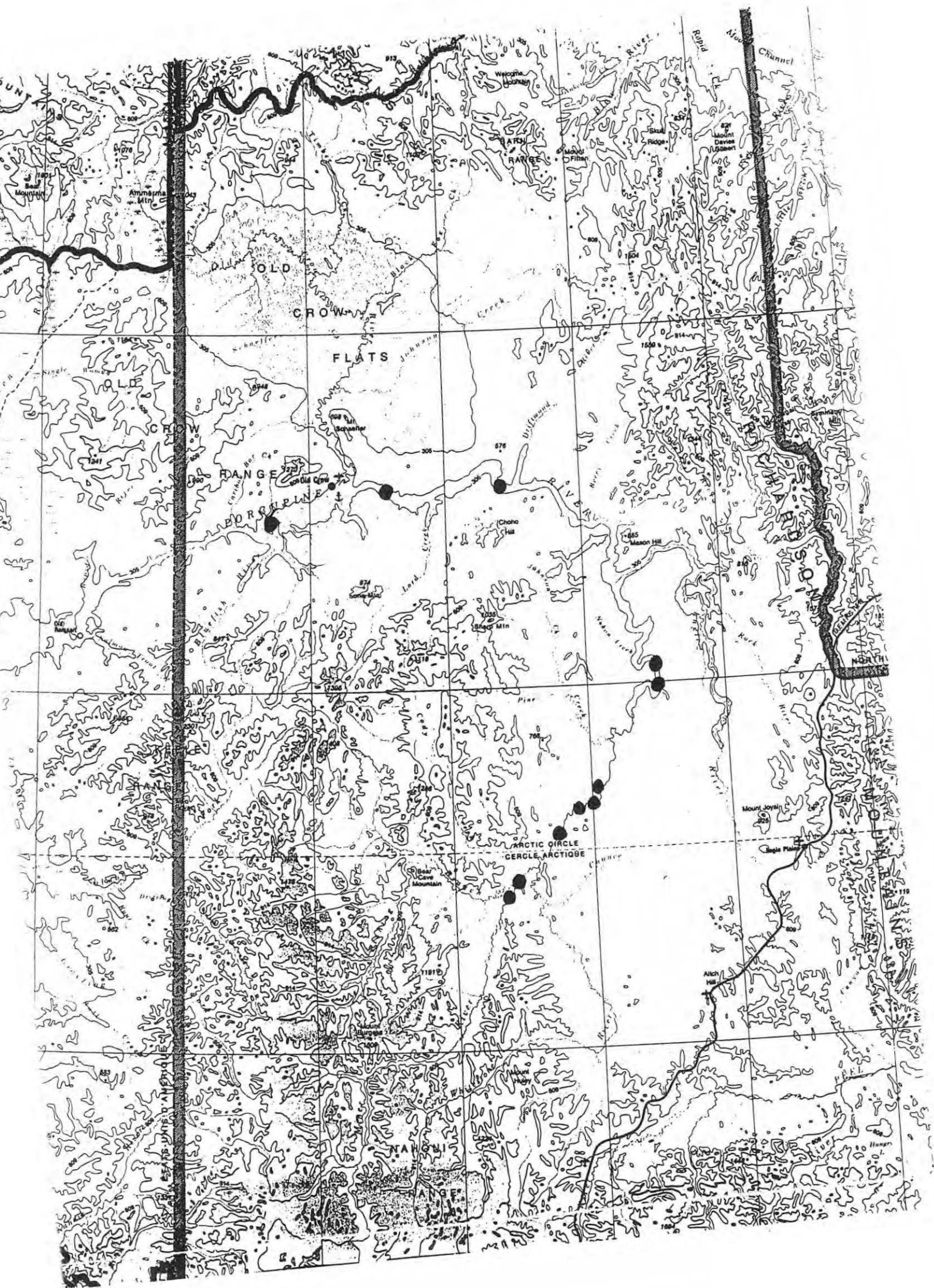


Figure 2. Porcupine River drainage in Canada. Locations at which radio tags were detected are marked ("●"). (Map - Yukon Territory 1:1,000,000.)



Photo 1. Maule M7 equipped with Yagi antennae.



Photo 2. Porcupine River in vicinity of Schaeffer Creek.



Photo 3. Porcupine River between Tizra Creek and Nukon Creek. Radio signal and live fish observed here.

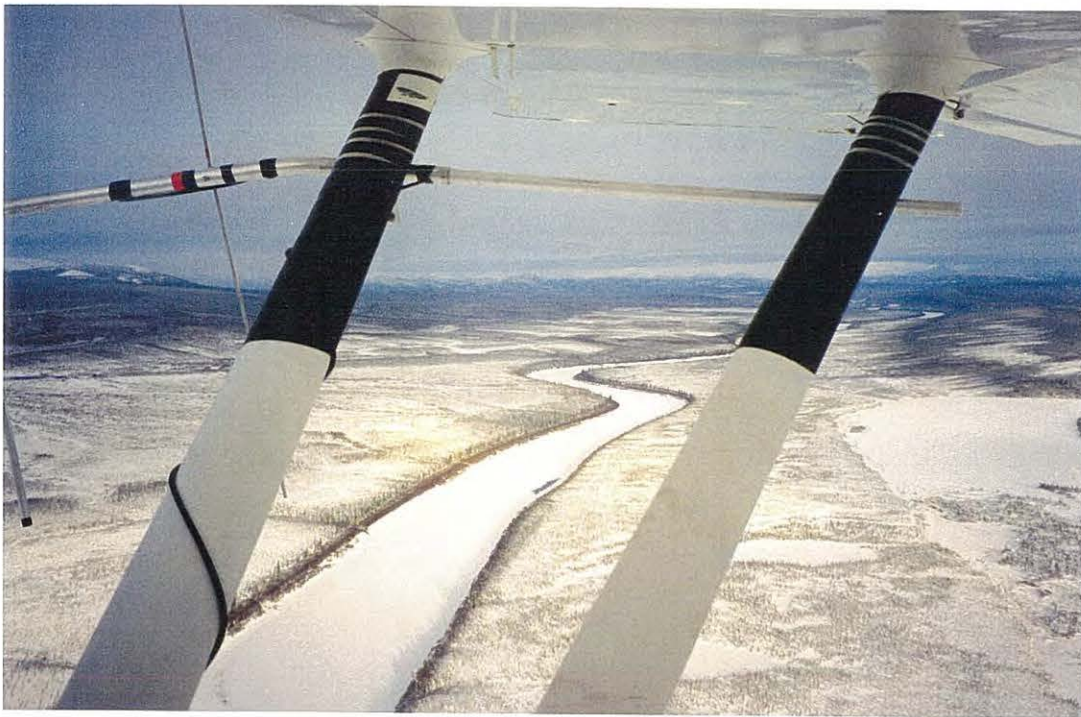


Photo 4. Bell River near confluence with Porcupine River.

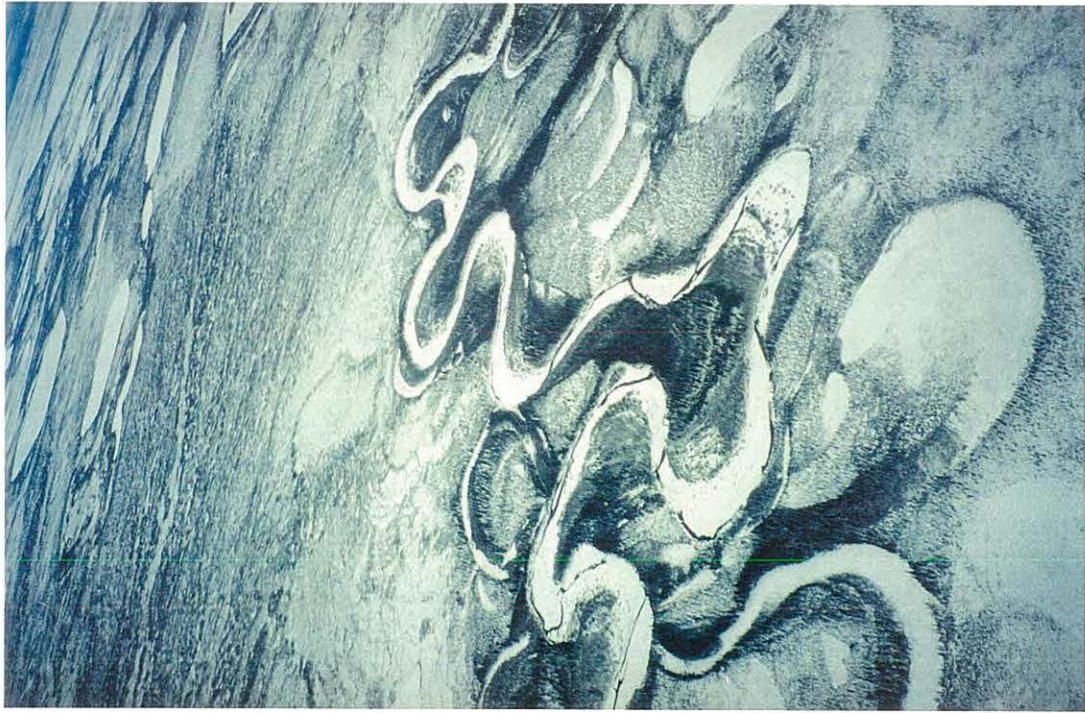


Photo 5. Black Fox Creek at the location where open water was observed.



Photo 6. Spawning area on the Fishing Branch River at Bear Cave Mountain. Note open water. Weir site and tracking station are a short distance downstream from this section of river.



Photo 7. Remote tracking station on the Fishing Branch River at Bear Cave Mountain. Weir is visible in the background.